

Akhilesh Raj P

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SUMMARY

Computer Science and Engineering undergraduate with hands-on experience in full-stack web development, embedded IoT systems, and AI-powered application development. Proficient in React.js, Node.js, Express.js, Python, and MongoDB. Completed a Full Stack Developer internship at Webbitech. Passionate about building real-world solutions at the intersection of AI, IoT, and software engineering.

EDUCATION

Sri Ramakrishna Engineering College, Coimbatore 2024 - 2028
B.E. Computer Science and Engineering (Anna University) | CGPA: 6.98 / 10 | No Arrears
Bharathi Matriculation Higher Secondary School, Coimbatore HSC - State Board | 2024 | 77.67%
Amrita Vidyalaya, Coimbatore SSLC - CBSE | 2022 | 84%

EXPERIENCE

Full Stack Developer Intern - Webbitech Coimbatore | 2026 (15 Days)

- Developed and delivered full-stack web application projects using React.js for frontend and Node.js with Express.js for backend REST API development, applying industry-standard coding practices.
- Designed and implemented reusable UI components, integrated frontend interfaces with backend API endpoints, and performed API testing to ensure reliable data flow between layers.
- Maintained organized project structure and managed source code using Git and GitHub for version control and timely submission of all deliverables.

Tech: React.js, Node.js, Express.js, HTML5, CSS3, JavaScript, REST APIs, Git, GitHub

TECHNICAL SKILLS

Languages: C, C++, Java, Python, SQL
Core Strengths: Problem Solving, Full-Stack Development, Rapid Prototyping
Web & Backend: React.js, Node.js, Express.js, FastAPI, Tailwind CSS, JavaScript, REST APIs, GitHub / Slack / Twilio / WhatsApp APIs
Databases: Relational Databases, SQLite, PostgreSQL, Firebase, MongoDB
Tools: Git, GitHub, VS Code, Postman, Arduino IDE
AI / ML: LLaMA (Ollama), Groq API, Speech-to-Text, NLP, Prompt Engineering
Embedded / IoT: Arduino, ESP32, IR Sensor, HC-SR04, 433 MHz RF

PROJECTS

KAVACH - Automatic Train Protection System Mini Project I | April 2026
Tech: Arduino Nano, ESP32, IR Sensor, HC-SR04 Ultrasonic, SG90 Servo, 433 MHz RF, L298N Motor Driver, C/C++, HTML/CSS/JS

- Designed a dual-module embedded IoT safety system — Train Safety Monitoring Unit and Automatic Gate Control Unit — communicating via 433 MHz RF; achieved 97.2% obstacle detection accuracy and 97.5% overall system efficiency across 100 test scenarios.
- Implemented IR proximity-based obstacle detection with interrupt-driven stepwise motor deceleration; built a 4-state automated gate control state machine with SG90 servo motor, buzzer, and LED visual alerts.
- Integrated ESP32 WiFi module for a real-time HTTP monitoring dashboard displaying live train status, gate position, and sensor readings. Prototype cost approximately Rs. 3,000-4,000.
- Awarded Best Mini Project at Sri Ramakrishna Engineering College; presented at INTELEC 2026 National Level Technical Symposium at KPRIET, organized by IEEE.

ShiftAI - AI-Powered Industrial Shift Handover System 2025
Tech: React.js, Vite, Node.js, Express.js, Groq API, LLaMA Models, Twilio WhatsApp API, MongoDB, Bootstrap

- Built a full-stack voice-based industrial reporting platform that converts supervisor speech into structured operational intelligence using LLaMA AI models via Groq API — enabling automatic issue extraction and incident classification (Mechanical, Electrical, Safety).
- Integrated Twilio WhatsApp API for real-time manager notifications; implemented JWT-based role access control (Supervisor, Manager, Admin), analytics dashboard with trend analysis, and export support in PDF, CSV, and Excel formats.

ACHIEVEMENTS

- Awarded Best Mini Project at Sri Ramakrishna Engineering College for KAVACH - Automatic Train Protection System.
- Presented KAVACH prototype before academic experts and industry evaluators at ProjectXpress - Project Expo, KPRIET (IEEE INTELEC 2026 National Level Technical Symposium).